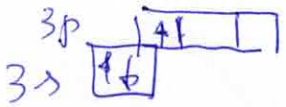
1

a) Al $Z = 13$ $\quad 1s^2 2s^2 2p^6 3s^2 3p^1$

riempimento tutte shell e sotto shell

1 solo e^- spaiato (di valenza)



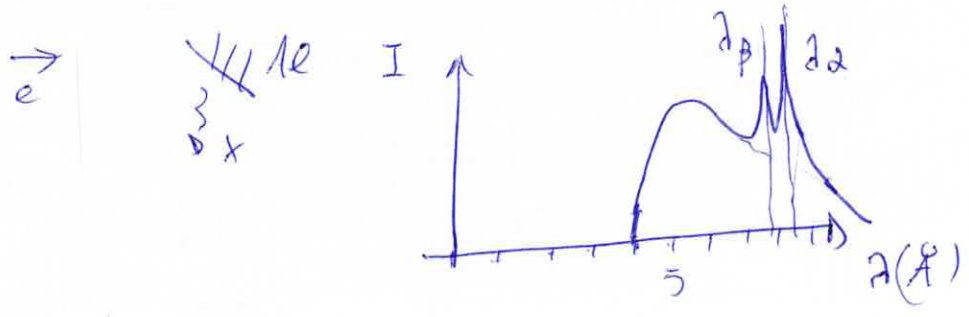
$s_{tot} = 1/2 \qquad 2s+1 = 2$

$l_{tot} = 1 \quad \Rightarrow P$

$j_{tot} = 1/2,\ 3/2$

stato fondamentale j_{tot} minimo $\Rightarrow$ $\boxed{{}^2P_{1/2}}$

b)



soglia λ minima

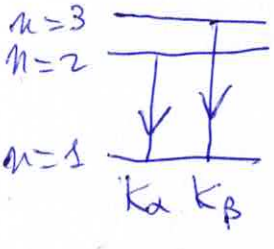
$e\Delta V = h\nu_{max} = h\dfrac{c}{\lambda_{min}}$

$\rightarrow \lambda_{min} = \dfrac{hc}{e\Delta V} = \dfrac{6.6\,10^{-34}\ 3\,10^8}{1.6\,10^{-19}\ 3\,10^3} = \dfrac{20\ 10^{-26}}{5\ 10^{-16}} \sim 4\,10^{-10}\,m = 4\,\text{Å}$

$\left(\dfrac{1.24}{E(eV)}\,\mu m = 4.1\,\text{Å} \right) \qquad \lambda(\text{Å}) \simeq 12375/E(eV)$

$\rightarrow$ background: radiazione frenamento

$\rightarrow K_\alpha, K_\beta$ riempimento lacuna shell K



$h\nu_\beta > h\nu_\alpha \Rightarrow \lambda_\beta < \lambda_\alpha$

$K_\alpha : n=2 \rightarrow n=1$

$K_\beta : n=3 \rightarrow n=1$

$h\nu_\alpha = E_{n=2} - E_{n=1}$ $\qquad E_n = -\frac{Z_n}{n^2}$

c) con $Z_1 \approx Z-2 = 11$

$Z_2 \approx Z-10 \approx 3$

$Z_3 \approx n \approx 3 \times 3$

$$E_{n=1} \simeq -11^2 \times 13.6\ eV \sim 1646\ eV$$

$$E_{n=2} \simeq -3^2 \frac{13.6}{4} \sim 30.6\ eV$$

$$E_{n=3} \sim -3^2 \frac{13.6}{9} \sim 13.6\ eV$$

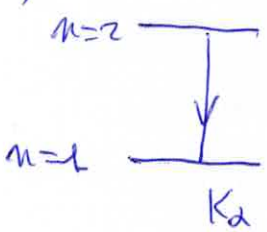
$$h\nu_\alpha = 1646 - 31 = 1615\ eV \Rightarrow \lambda_\alpha = 7.67\ Å$$

$$h\nu_\beta = 1646 - 14 \simeq 1632\ eV \Rightarrow \lambda_\beta = 7.58\ Å$$

$$\lambda_\alpha^{calc} = 7.67 \qquad \lambda_\alpha^{sper} = 8.34 \qquad \Delta\lambda/\lambda \sim \frac{0.6}{8} \sim 10\%$$

$$\lambda_K \sim \frac{1.24}{1646} 10^4 = 7.52\ Å$$ usando solo $E_K = (Z-2)^2 \times 13.6\ eV$

d)

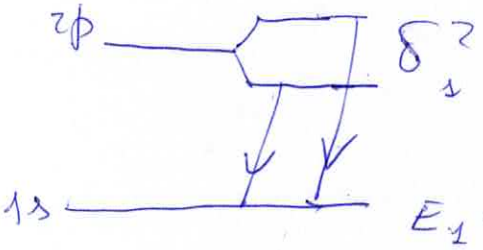


me radiative $\Rightarrow \Delta l = \pm 1$

$\Rightarrow$ parte da 2p

lacune in 2p $\Rightarrow$ come 1 solo e^-

$\Rightarrow j = 1/2,\ 3/2$



$\delta_{S.O.} = 0.02\ Å \Rightarrow$

$E_1 \sim 1485.59 \qquad E_2 \sim 1482.0 \Rightarrow \delta E_{S.O} = 3.6\ eV$

$$\frac{\Delta E}{E} = \frac{\Delta\lambda}{\lambda} \Rightarrow \Delta E = E \cdot \frac{\Delta\lambda}{\lambda} = 1640 \times \frac{0.02}{8} = 4.1\ eV$$

(2)

$M, q, s = 1$ ferma $\Rightarrow \vec{L} = 0$ $\quad l = 0$

$\vec{J} = \vec{S}$ $\quad j = s = 1$

$\hat{J}^2 \rightarrow$ autovalori $j(j+1)\hbar^2$ $\quad m_j = -j \dots +j$

$\hat{J}_z \rightarrow$ autovalori $m_j \hbar$ $\quad = -1, 0, +1$

operatori matrici 3×3

a) $\hat{J}^2 = 1(1+1)\hbar^2 \begin{pmatrix} 1 & & 0 \\ & 1 & \\ 0 & & 1 \end{pmatrix} = 2\hbar^2 \hat{1}$

$\hat{J}_z \rightarrow \hbar \begin{pmatrix} 1 & & \\ & 0 & \\ & & -1 \end{pmatrix}$ diagonali

autovalori $\rightarrow$ possibili risultati misura

b) $\hat{H} = ?$ interazione $U = -\vec{\mu} \cdot \vec{B} = -\mu_z B$

$\hat{H} = -(\vec{\mu}_j)_z B = -\left(-g_j \frac{\mu_B}{\hbar} J_z\right) B = g_j \frac{qB}{2m} \hat{J}_z$

$g_j = g_s = 2$ $\quad \hat{H} = \frac{qB}{M} \hat{J}_z$

$\hat{H} = \frac{qB}{M} \hbar \begin{pmatrix} 1 & & \\ & 0 & \\ & & -1 \end{pmatrix}$ diagonale

$|\alpha(t=0)\rangle = A\,|1,1\rangle + B\,|1,0\rangle + C\,|1,-1\rangle$

autoket $|j, m_j\rangle \rightarrow |1,1\rangle, |1,0\rangle \quad |1,-1\rangle$

$\hat{J}^2 |j, m_j\rangle = j(j+1)\hbar^2 |j, m_j\rangle$ $\quad \hat{J}_z |j, m_j\rangle = m_j \hbar |j, m_j\rangle$

con $|A|^2 + |B|^2 + |C|^2 = 1$

$t=0$ $\quad \langle J^2 \rangle = 2\hbar^2$

$$\langle J_z \rangle = \left(|A|^2 - |C|^2\right)\hbar$$

$$\langle \alpha | \hat{J}_z | \alpha \rangle = \langle \alpha | \left(\hbar |1,1\rangle A + 0 |1,0\rangle B - \hbar |1,-1\rangle C \right)$$

$$= \hbar |A|^2 \cdot 1 + |C|^2 (-\hbar)$$

per $\hat{H}$: ~~$\hat{H}|\alpha\rangle = \hat{H}$~~ autovalori

$$E_+ = \frac{e\hbar}{M} B \qquad E_0 = 0 \qquad E_- = -\frac{e\hbar B}{M}$$

$$\langle \hat{H} \rangle = \left(|A|^2 - |C|^2\right) \frac{e\hbar B}{M}$$

per $t > 0$

$$|\alpha(t)\rangle = A|1,1\rangle e^{-iE_+t/\hbar} + B|1,0\rangle \cdot 1 + C|1,-1\rangle e^{-iE_-t/\hbar}$$

ma operatori commutano tutti con $\hat{H}$

$\Rightarrow$ valor medi <u>non</u> variano nel tempo

$$\langle \hat{J}^2 \rangle_t = \langle \hat{J}^2 \rangle_0 \; ; \; \langle \hat{H} \rangle_t = \langle \hat{H} \rangle_0 \; ; \; \langle \hat{J}_z \rangle_t = \langle \hat{J}_z \rangle_0$$

d) $$\frac{d\langle \hat{J}_x \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{J}_x, \hat{H}] \rangle = \frac{1}{i\hbar} \langle \frac{eB}{M} [\hat{J}_x, \hat{J}_z] \rangle =$$

$x\,y\,z$
$z\,x\,y$
$y\,z\,x$

$$= \frac{1}{i\hbar} \langle \frac{eB}{M} (-i\hbar \hat{J}_y) \rangle =$$

$$= -\frac{eB}{M} \langle \hat{J}_y \rangle$$

$$\frac{d\langle \hat{J}_y \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{J}_y, \hat{H}] \rangle = \frac{1}{i\hbar} \langle \frac{eB}{M} [\hat{J}_y, \hat{J}_z] \rangle =$$

$$= \frac{1}{i\hbar} \frac{eB}{M} \langle i\hbar \hat{J}_x \rangle = \frac{eB}{M} \langle \hat{J}_x \rangle$$

$$\frac{d^2\langle \hat{J}_x\rangle}{dt} = -\frac{eB}{M}\frac{d\langle J_y\rangle}{dt} = -\frac{eB}{M}\frac{qB}{M}\langle J_x\rangle = -\omega^2\langle \hat{J}_x\rangle$$

$$\Rightarrow \langle \hat{J}_x\rangle \sim \cos\omega t \quad \text{con} \quad \omega = \frac{qB}{M}$$

stesso cose per $\langle \hat{J}_y\rangle$!